ONLINE FOOD DELIVERY BUSINESS ANALYTICS SYSTEM

PROJECT REPORT

1. ***INTRODUCTION***

The **Online Food Delivery Business Intelligence & Performance Analysis Using SQL** project focuses on analyzing transactional and operational data from an online food delivery platform to extract meaningful business insights. The dataset consists of multiple interconnected entities including customers, restaurants, orders, order items, and delivery agents, representing the complete lifecycle of a food delivery transaction. By leveraging this relational schema, the project aims to evaluate business performance, customer behavior, restaurant efficiency, and delivery operations through structured data analysis.

Analyzing such data is crucial for understanding revenue trends, identifying high-performing restaurants and cuisines, improving delivery efficiency, and enhancing customer satisfaction. With increasing competition in the food delivery industry, data-driven decision-making plays a vital role in optimizing operational processes, reducing delivery delays, improving customer retention, and maximizing profitability. This project highlights how structured business data can be transformed into actionable insights that support strategic planning and operational improvements.

Structured Query Language (SQL) was used as the primary analytical tool to perform data extraction, aggregation, filtering, and multi-table joins across the relational database. Various SQL techniques such as aggregate functions, grouping, subqueries, and date-based analysis were applied to measure key performance indicators including revenue growth, customer ordering patterns, discount impact, and delivery performance. The project demonstrates real-world relevance by simulating analytical scenarios commonly encountered in food delivery platforms, thereby showcasing practical SQL skills and business intelligence capabilities essential for data analyst roles.

2. ***PROBLEM STATEMENT***

The rapid growth of online food delivery platforms has led to the generation of large volumes of transactional and operational data. However, without systematic analysis,

businesses struggle to fully utilize this data to monitor performance, understand customer behavior, optimize delivery operations, and identify revenue growth opportunities.

The absence of structured business intelligence can result in inefficient delivery processes, poor customer retention, suboptimal restaurant partnerships, and ineffective promotional strategies. Therefore, there is a need to analyze the available relational data to uncover insights that support data-driven decision-making and operational efficiency.

This project aims to address several critical business questions, including:

1. Which restaurants and cuisines generate the highest revenue?
2. How do customer demographics and city-based trends influence ordering behavior?
3. What is the impact of discounts on overall revenue and order frequency?
4. How efficient are delivery operations in terms of delivery time and agent performance?

Additionally, the project investigates payment preferences, repeat customer patterns, and item-level sales trends to better understand purchasing behavior and platform performance.

Solving these problems is essential for improving customer satisfaction, strengthening restaurant partnerships, optimizing delivery logistics, and maximizing profitability. By identifying performance gaps and growth opportunities, the business can enhance operational efficiency, reduce delivery delays, implement targeted marketing strategies, and improve customer retention. Ultimately, this analysis enables the platform to transition from reactive decision-making to proactive, insight-driven strategies that foster sustainable business growth.

3. BUSINESS OBJECTIVES

- Act as a Data Analyst to extract and analyze transactional and operational data using SQL in order to generate actionable business insights for strategic decision-making.
- **Measure and improve revenue performance** by identifying monthly revenue trends, average order value, city-wise revenue contribution, and the impact of discounts on overall profitability.
- **Enhance customer retention and engagement** by analyzing new versus returning customers, repeat purchase behavior, signup trends, and demographic-based ordering patterns.

- **Evaluate restaurant growth and performance** by identifying top-performing restaurants, cuisine popularity, rating-based revenue patterns, and city-level restaurant demand.
- **Analyze product-level performance** by determining most frequently ordered items, high-revenue generating products, basket size trends, and item-level sales contribution.
- **Improve delivery and operational efficiency** by monitoring average delivery time, identifying city-wise delivery delays, and evaluating delivery agent performance through ratings and workload distribution.
- **Understand customer payment behavior** by measuring payment method preferences and their relationship with order value and order frequency.
- **Support data-driven decision-making** by generating SQL-based KPIs and insights that help optimize marketing strategies, restaurant partnerships, discount strategies, and overall platform performance.

4. DATASET OVERVIEW

4.1 Dataset Description

The dataset represents a structured relational database of an online food delivery platform capturing customer information, restaurant details, order transactions, item-level purchases, and delivery agent data. It reflects the complete operational workflow of the platform — from customer registration and order placement to restaurant fulfillment and delivery execution. The dataset is designed to support business intelligence and performance analysis through SQL-based querying and aggregation across multiple entities.

4.2 Tables Involved

The analysis is performed using five primary tables:

- **Customers:** Contains demographic and registration details of users including location and signup information.
- **Restaurants:** Stores restaurant attributes such as cuisine type, city, and ratings.
- **Orders:** Represents transactional order data including order amount, discounts, payment method, and delivery time.
- **Order_Item:** Provides item-level order breakdown including product name, quantity, and pricing for granular sales analysis.

- **Delivery Agents:** Maintains delivery personnel details including city, joining date, and performance ratings.

These tables are interconnected through primary and foreign keys, enabling multi-dimensional analysis using SQL joins.

4.3 Record Size

The dataset consists of multi-table transactional data where the **Orders** and **Order_Item** tables typically contain the highest volume of records due to frequent order activity, while **Customers**, **Restaurants**, and **Delivery Agents** represent relatively stable master data. This combination reflects both high-frequency transactional data and slowly changing dimensional data.

4.4 Time Period

The dataset includes temporal attributes such as **signup_date**, **order_date**, and **joining_date**, enabling time-based analysis of customer growth, revenue trends, delivery performance, and operational expansion over a continuous business period.

4.5 Type of Data Stored

The dataset contains a mix of:

- **Demographic data:** Customer gender, city, and registration details
- **Operational data:** Order transactions, delivery time, and agent performance
- **Financial data:** Order amount, discounts, and pricing information
- **Categorical data:** Cuisine types, payment methods, and item names
- **Performance metrics:** Restaurant ratings and delivery agent ratings

This diverse data structure enables comprehensive analysis across revenue, customer behavior, restaurant performance, and delivery operations, making the dataset suitable for end-to-end business intelligence reporting.

5. ***ER DIAGRAM EXPLANATION***

The Entity Relationship (ER) Diagram of the **Online Food Delivery Business Intelligence & Performance Analysis** database represents the structural design of the system by illustrating how different entities interact to support the end-to-end food ordering and delivery process. The diagram visually depicts the flow of data between customers, restaurants, orders, order items, and delivery agents, ensuring data integrity and enabling efficient analytical querying.

5.1 Entities

The database consists of five primary entities:

- **Customers:** Represents platform users who place food orders. This entity stores demographic and registration details used for customer behavior and retention analysis.
- **Restaurants:** Represents partner restaurants providing food services. It includes attributes such as cuisine type, location, and ratings for performance evaluation.
- **Orders:** Acts as the central transactional entity capturing each order placed by customers, including order value, discount, payment method, and delivery duration.
- **Order_Item:** Represents item-level details within each order, allowing granular tracking of product sales, quantity, and pricing.
- **Delivery Agents:** Represents delivery personnel responsible for fulfilling orders, including their location, joining date, and performance ratings.

5.2 Relationships

The entities are connected through logical relationships:

- A **Customer places Orders** — linking customer behavior with transactional activity.
- A **Restaurant receives Orders** — enabling restaurant-level revenue and performance analysis.
- An **Order contains multiple Order Items** — representing the breakdown of each order into individual food products.
- Delivery Agents operate independently in the schema but are associated with operational delivery performance across cities.

5.3 Cardinality

The ER diagram follows these cardinality rules:

- **Customers → Orders:** One-to-Many (1:N)
    - One customer can place multiple orders, but each order belongs to a single customer.
- **Restaurants → Orders:** One-to-Many (1:N)
    - A restaurant can receive many orders, while each order is associated with only one restaurant.

- **Orders → Order_Item:** One-to-Many (1:N)
    - Each order can contain multiple items, whereas each order item belongs to a single order.
- **Delivery Agents:** Represent an independent entity with potential implicit relationships to operational metrics rather than direct transactional linkage.

5.4 Key Constraints

The database enforces data integrity through primary and foreign key constraints:

- **Primary Keys**
    - customer_id in Customers
    - restaurant_id in Restaurants
    - order_id in Orders
    - order_item_id in Order_Item
    - agent_id in Delivery Agents
- **Foreign Keys**
    - customer_id in Orders references Customers
    - restaurant_id in Orders references Restaurants
    - order_id in Order_Item references Orders

These constraints ensure referential integrity, prevent orphan records, and enable accurate multi-table joins required for analytical queries.

6. ***EXPLORATORY DATA ANALYSIS (EDA)***

Exploratory Data Analysis (EDA) was conducted to understand the structure, quality, and distribution of the data before performing business intelligence queries. The objective of EDA was to identify data patterns, detect inconsistencies, evaluate relationships between entities, and prepare the dataset for accurate analytical reporting. Using SQL queries, the dataset was examined across customers, restaurants, orders, order items, and delivery agents to gain initial insights into business activity and data completeness.

6.1 Data Structure & Integrity Checks

Initial exploration involved validating table structures, primary keys, and relationships between entities. Record counts were analyzed across tables to understand data

volume, ensuring that transactional tables such as Orders and Order_Item contained consistent mappings with Customers and Restaurants. Checks for missing values, duplicate records, and invalid entries were performed to maintain data accuracy and prevent analytical bias.

6.2 Customer Exploration

Customer data was analyzed to understand demographic distribution, signup trends, and geographic presence. City-wise customer counts helped identify high-demand markets, while gender-based distribution supported demographic segmentation. Signup date analysis provided insight into customer acquisition trends and platform growth over time.

6.3 Restaurant Exploration

Restaurant-level exploration focused on cuisine diversity, city distribution, and rating patterns. This helped identify dominant cuisines, concentration of restaurants in specific locations, and variations in restaurant ratings that could influence customer ordering behavior.

6.4 Order & Revenue Exploration

The Orders table was explored to understand order frequency, revenue distribution, discount usage, and payment method patterns. Time-based exploration of order_date revealed peak ordering periods, while analysis of order_amount and discount provided an early understanding of revenue variability and promotional impact.

6.5 Product-Level Exploration

Item-level exploration from the Order_Item table helped identify commonly ordered food items, quantity patterns, and pricing variations. This analysis provided a foundation for understanding customer basket composition and detecting potential high-revenue menu items.

6.6 Delivery Performance Exploration

Delivery agent data and delivery_time were analyzed to assess operational efficiency. City-wise agent distribution, agent ratings, and delivery duration variability were explored to detect potential service bottlenecks and performance inconsistencies.

6.7 EDA Outcome

The exploratory analysis provided a clear understanding of data distribution. It also ensured data readiness for advanced SQL analysis by highlighting key dimensions such as customer behavior, restaurant demand, revenue drivers, and delivery performance. These findings formed the basis for subsequent KPI calculations, trend analysis, and business insight generation.

7. ***SQL Query Explanation – Total Revenue Calculation***

SELECT SUM(order_amount - discount) AS total_revenue

FROM orders;

7.1 Business Question

What is the **total net revenue generated by the platform** after accounting for discounts offered on all customer orders?

This question helps determine the actual earnings of the business rather than the gross order value.

7.2 Tables Used

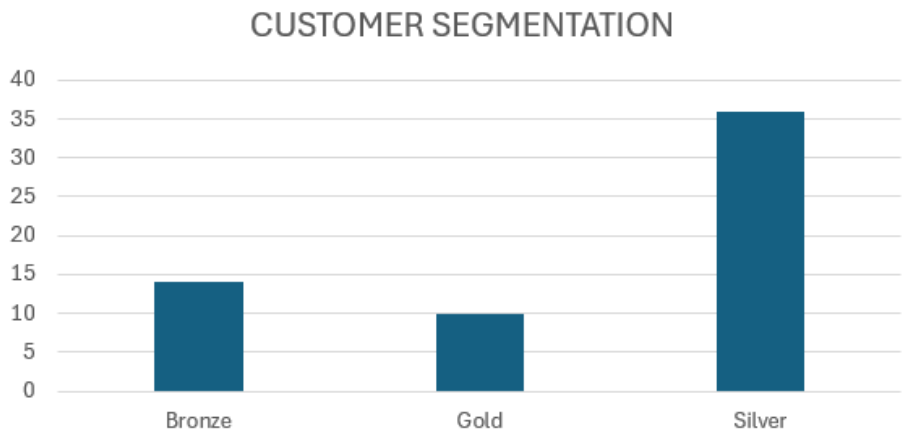
- **Orders Table:**
  Contains transactional data including order_amount and discount. The query uses this table to compute revenue after subtracting promotional discounts applied to each order.

7.3 Business Insight Derived

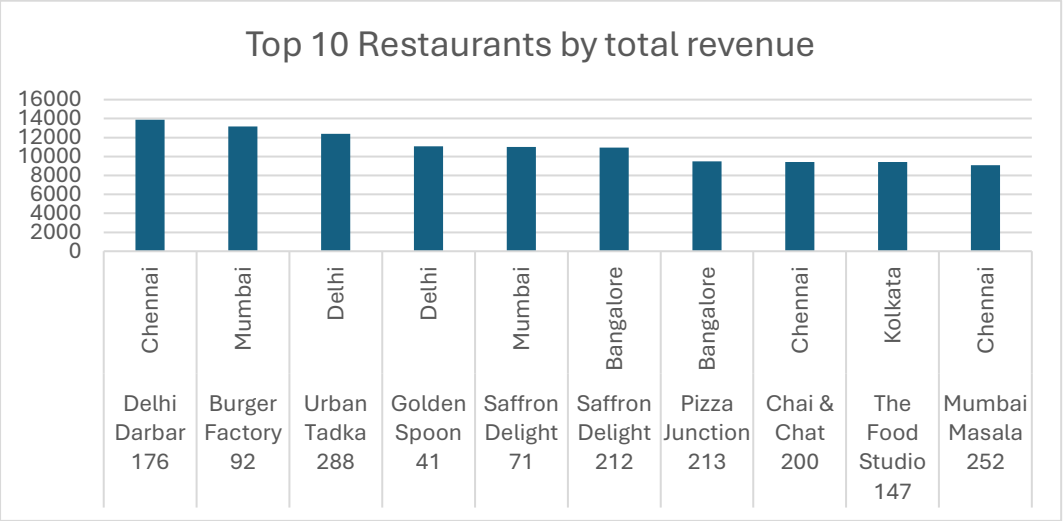
- Provides the **true revenue figure** reflecting the company's earnings after discount expenses.
- Helps evaluate whether discount strategies are sustainable.
- Supports financial reporting and profitability analysis.
- Enables comparison with gross revenue to measure promotional impact.

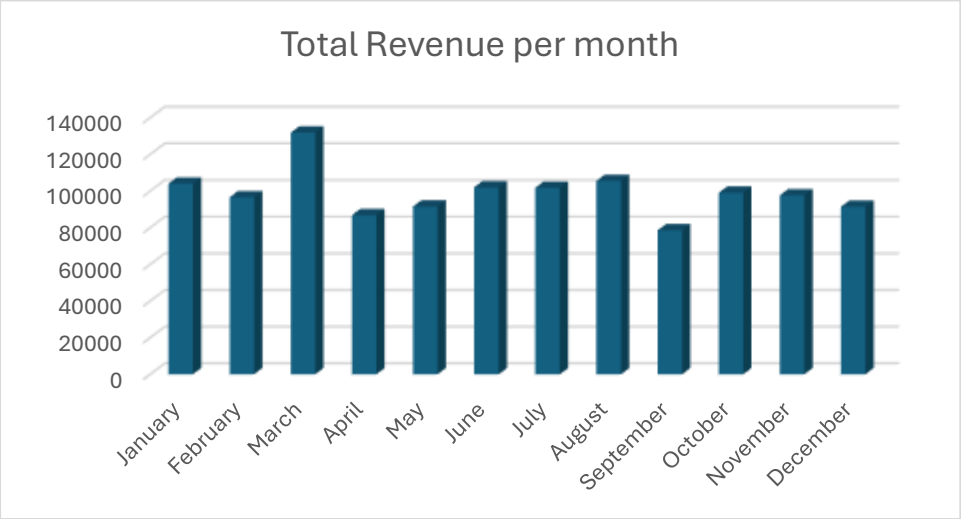
8. ***DATA ANALYSIS AND INSIGHTS***

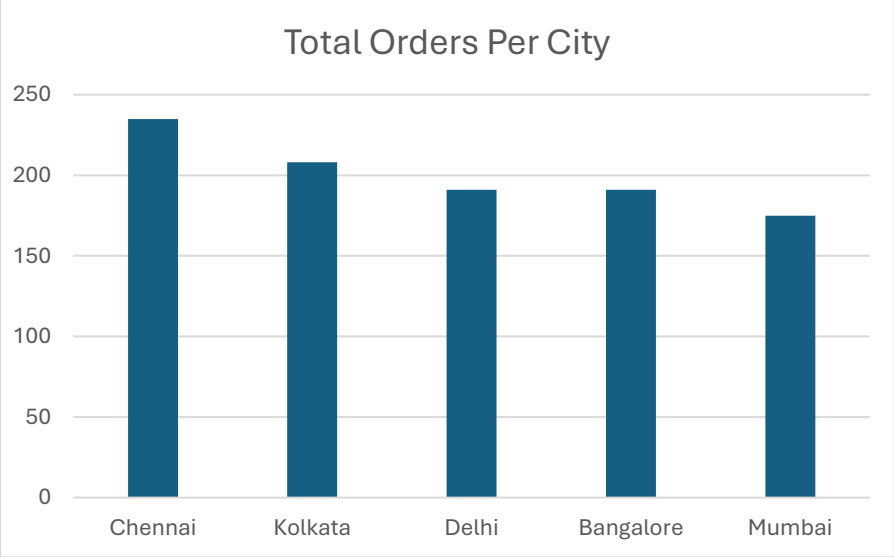
8.1 CUSTOMER SEGMENTATION



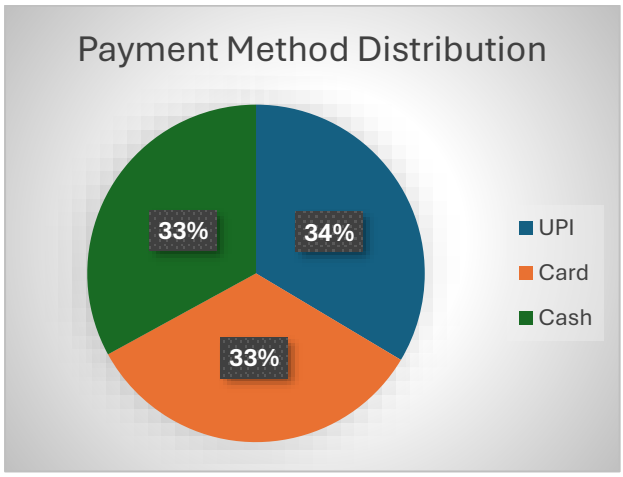
8.2 RESTAURANT PERFORMANCE ANALYSIS



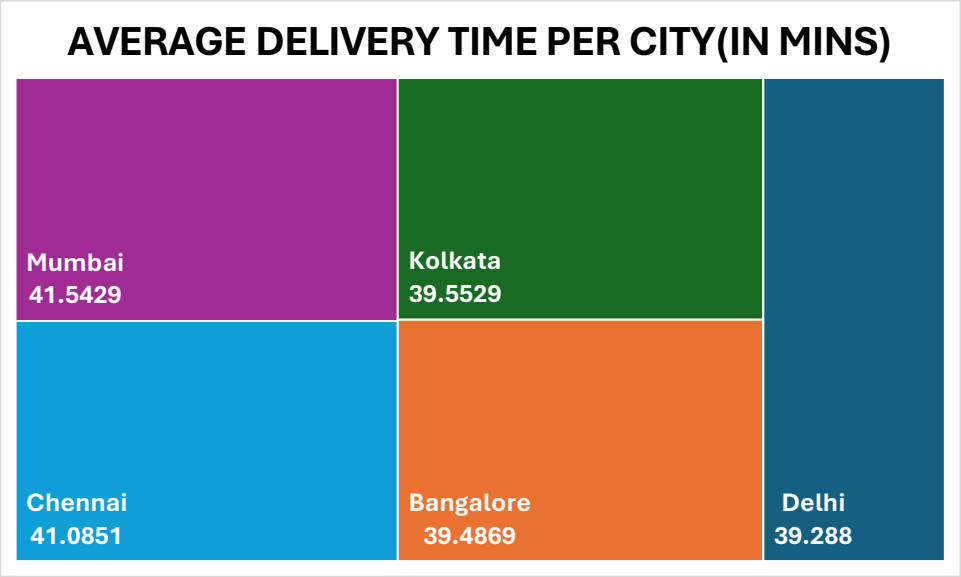


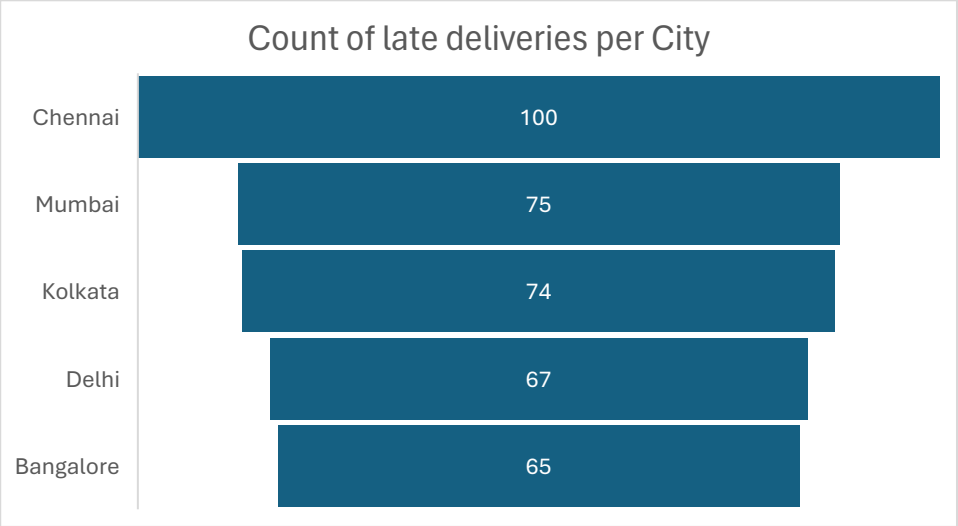


8.3 CUSTOMER PAYMENT DISTRIBUTION



8.4 DELIVERY PERFORMANCE ANALYSIS





Based on the SQL analysis performed across customers, restaurants, orders, order items, and delivery agents, several meaningful business insights were identified. The visualizations presented above highlight key operational patterns, revenue distribution, and customer behavior trends that are critical for strategic decision-making.

9. ***VIEWS IMPLEMENTATION***

Purpose

A **View** is a virtual table created from a SQL query that simplifies complex joins and aggregation logic. In this project, the view was created to calculate restaurant-level revenue, enabling reusable and simplified reporting without repeatedly writing aggregation queries.

Business Value

- Simplifies revenue reporting
- Improves query readability
- Enables dashboard integration
- Enhances data abstraction and security

Query Used

```
CREATE VIEW restaurant_revenue_view AS
SELECT r.restaurant_name, r.city,
    SUM(o.order_amount - o.discount) AS total_revenue
FROM orders o
JOIN restaurants r ON o.restaurant_id = r.restaurant_id
GROUP BY r.restaurant_name, r.city;
```

This view calculates **net revenue per restaurant and city**, making it easier to perform ranking and performance analysis.

10. ***STORED PROCEDURE IMPLEMENTATION***

Purpose

A **Stored Procedure** encapsulates reusable business logic that can be executed with parameters. It improves modularity, reduces query repetition, and enhances security.

Business Value

- Dynamic retrieval of top-performing restaurants

- Faster execution of repeated analytical queries
- Supports dashboard and reporting automation

Query Used

```
CREATE PROCEDURE GetTopRestaurants(IN top_n INT)
BEGIN
  SELECT restaurant_name, city, total_revenue
  FROM restaurant_revenue_view
  ORDER BY total_revenue DESC
  LIMIT top_n;
END;
```

Execution

```
CALL GetTopRestaurants(5);
```

This procedure dynamically returns the **Top N restaurants by revenue**, enabling flexible performance analysis.

11. INDEXING STARTEGY

Purpose

Indexes were implemented to reduce query execution time and improve filtering and join efficiency on frequently accessed columns.

Indexes Created

Index on order_date

Supports faster time-based filtering and trend analysis.

```
CREATE INDEX idx_order_date ON orders(order_date);
```

Index on customer name

Improves customer search and segmentation queries.

```
CREATE INDEX idx_customer_name ON customers(name);
```

Performance Benefits

- Reduces full table scans
- Accelerates joins and filtering
- Improves reporting query performance and supports scalability

12. TRIGGER IMPLEMENTATION

Triggers were implemented to automate monitoring, enforce data integrity, and capture operational exceptions.

High-Value Order Logging Trigger

Purpose

Automatically logs orders exceeding a defined threshold (₹1000), helping identify premium customers and high-value transactions.

```
create table high_value_orders_log(
  log_id int primary key auto_increment,
  order_id int,
  customer_id int,
  restaurant_id int,
  order_amount decimal(10,2),
  log_date datetime default current_timestamp
);

CREATE TRIGGER trg_high_value_order
AFTER INSERT ON orders
FOR EACH ROW
BEGIN
  IF NEW.order_amount > 1000 THEN
    INSERT INTO high_value_orders_log (order_id, customer_id, restaurant_id,
order_amount)
    VALUES (NEW.order_id, NEW.customer_id, NEW.restaurant_id,
NEW.order_amount);
  END IF;
END;
```

Negative Discount Prevention Trigger

Purpose

Ensures data integrity by preventing invalid negative discount values.

```
CREATE TRIGGER trg_prevent_negative_discount
BEFORE INSERT ON orders
FOR EACH ROW
BEGIN
  IF NEW.discount < 0 THEN
    SET NEW.discount = 0;
  END IF; END;
```

Delivery Delay Monitoring Trigger

Purpose

Logs orders with delivery time exceeding 45 minutes to track operational delays.

```
create table delivery_delay_log(
  log_id int primary key auto_increment,
  order_id int,
  customer_id int,
  restaurant_id int,
  delivery_time int,
  created_at datetime default current_timestamp
);
CREATE TRIGGER trg_delivery_delay_warning
AFTER INSERT ON orders
FOR EACH ROW
BEGIN
  IF NEW.delivery_time > 45 THEN
    INSERT INTO delivery_delay_log (order_id, customer_id, restaurant_id,
delivery_time)
    VALUES (NEW.order_id, NEW.customer_id, NEW.restaurant_id, NEW.delivery_time);
  END IF;
END;
```

The implementation of views, stored procedures, indexing, and triggers enhances:

- Query performance and scalability
- Reusability of business logic
- Automated monitoring of critical events
- Data integrity and operational visibility
- Real-time detection of high-value orders and delivery delays

These techniques reflect production-level database practices and significantly strengthen the project's technical depth and real-world applicability.

13. <u>PERFORMANCE OPTIMIZATION</u>

Efficient query execution and scalable database design are critical for handling high-volume transactional data in an online food delivery platform. Since the dataset involves multiple relational tables with frequent joins and aggregations, performance optimization techniques were applied to ensure faster query response time, reduced resource consumption, and improved analytical scalability.

13.1 Importance of Indexing

Indexing played a key role in improving data retrieval speed, particularly for frequently queried and joined columns. Primary keys were automatically indexed to maintain entity uniqueness, while additional indexing was beneficial on:

- Foreign keys such as **customer_id**, **restaurant_id**, and **order_id** to accelerate join operations
- **order_date** to support time-based filtering and trend analysis
- **city** columns to optimize location-based segmentation queries

Proper indexing reduced full table scans, significantly improving aggregation and filtering performance on large transactional tables like Orders and Order_Item.

13.2 Query Optimization Techniques Used

Several query optimization strategies were applied:

- Selecting only required columns instead of using SELECT * to minimize data transfer
- Using aggregate functions with appropriate GROUP BY clauses to avoid unnecessary intermediate computations
- Applying WHERE filters early to reduce dataset size before joins
- Avoiding redundant subqueries by restructuring queries into more efficient forms
- Using calculated expressions (e.g., net revenue) within aggregation instead of multiple query passes

These techniques improved execution efficiency and ensured better readability and maintainability of SQL queries.

13.3 Join Efficiency

Since the analysis required combining transactional and dimensional data, join optimization was essential:

- Proper use of **INNER JOIN** for mandatory relationships to reduce unnecessary records
- Using indexed join keys to avoid costly nested loop operations
- Limiting multi-table joins to only necessary entities depending on the business question
- Aggregating data before joining where possible to reduce row-level processing

This approach improved performance while maintaining analytical accuracy.

13.4 CTE and Temporary Table Usage

Common Table Expressions (CTEs) and temporary result sets were used to enhance query structure and efficiency:

- **CTEs** helped break complex analytical queries into modular and readable steps
- Enabled reuse of intermediate aggregated results without repeating calculations
- Simplified ranking, segmentation, and KPI-based analysis
- Temporary tables can be used in large-scale scenarios to store intermediate results and reduce repeated computation overhead

This improved both query clarity and execution performance for advanced analytical tasks.

13.5 Scalability Considerations

To ensure the system remains efficient as data volume grows, the following scalability practices were considered:

- Designing normalized tables to reduce redundancy and maintain data integrity
- Using indexing strategies that balance read performance with write overhead
- Structuring queries to support incremental and time-based analysis
- Preparing the schema for integration with BI tools and data warehousing solutions
- Ensuring that aggregation-heavy queries can be adapted into materialized views for large-scale production environments

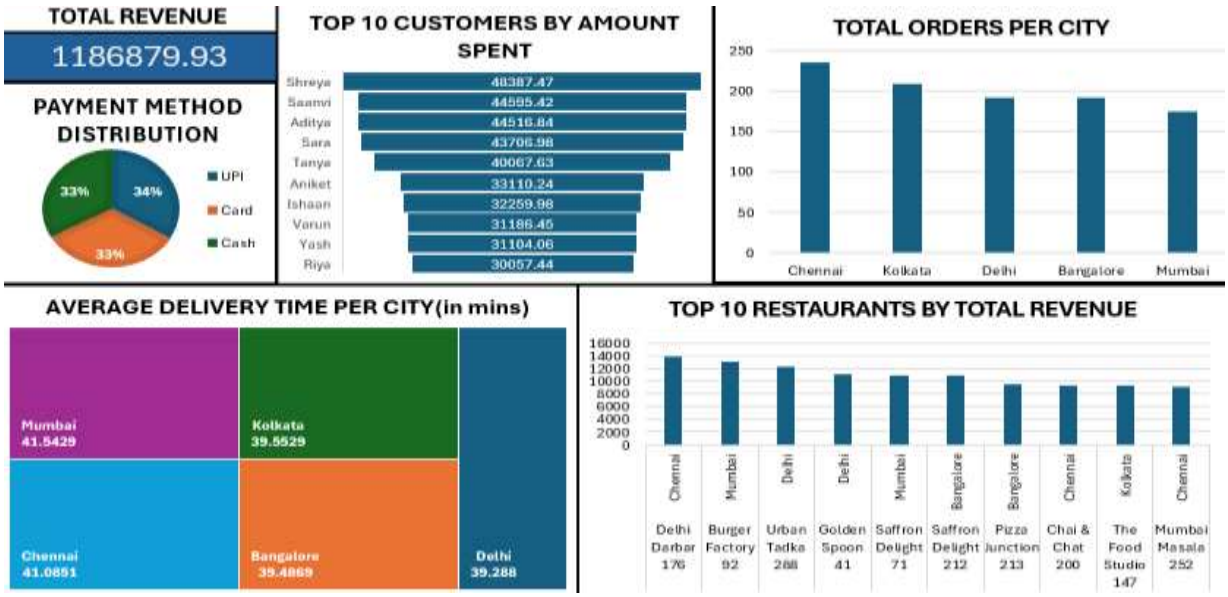
14. KEY FINDINGS SUMMARY

- **Strong Revenue Growth Trend:**
  The platform demonstrates consistent revenue generation driven by increasing order volume, indicating growing customer adoption and effective restaurant partnerships.
- **Repeat Customers Drive Significant Revenue:**
  Returning customers contribute a substantial share of total orders, highlighting strong customer loyalty and the importance of retention-focused strategies.
- **High Revenue Concentration Among Top Restaurants:**
  A limited number of top-performing restaurants generate a major portion of total

revenue, emphasizing their strategic importance to the platform's overall success.

- **Cuisine-Based Demand Patterns Identified:**
  Certain cuisines consistently attract higher order volumes, providing opportunities for targeted restaurant onboarding and menu expansion in high-demand categories.
- **Discounts Positively Influence Order Frequency:**
  Promotional discounts encourage customer purchases and boost short-term order volume, but require careful monitoring to maintain profitability.
- **Digital Payment Methods Dominate Customer Preference:**
  Customers show a strong inclination toward online and cashless payment methods, indicating trust in digital transactions and convenience-driven behavior.
- **Delivery Time Variability Impacts Customer Experience:**
  While average delivery times remain acceptable, inconsistencies across cities and delivery agents suggest opportunities for logistics optimization and workload balancing.
- **Product-Level Insights Reveal Key Revenue Drivers:**
  A small group of frequently ordered food items contributes significantly to total sales, enabling the business to focus on menu optimization, promotions, and bundling strategies.

15. <u>DASHBOARD AND VISUALIZATION</u>



16. ***BUSINESS RECOMMENDATIONS***

Based on the analysis of customer behavior, restaurant performance, order transactions, and delivery operations, the following actionable recommendations are proposed to improve business performance and ensure sustainable growth.

16.1 Revenue Growth

- **Strengthen partnerships with top-performing restaurants** by offering promotional visibility, exclusive deals, and priority listing to maximize revenue contribution from high-demand partners.
- **Expand high-demand cuisines** in cities showing strong ordering patterns to capture unmet customer demand and increase order volume.
- **Implement targeted discount strategies** instead of platform-wide promotions to balance customer acquisition with profitability.
- **Promote high-performing food items and combo offers** identified from item-level analysis to increase average order value and basket size.

16.2 Customer Retention

- **Introduce loyalty and reward programs** for repeat customers to reinforce engagement and encourage consistent ordering behavior.
- **Use personalized promotions** based on customer order history, city, and preferences to improve conversion and retention rates.
- **Improve onboarding incentives for new users** to convert first-time customers into repeat buyers more effectively.

16.3 Cost Efficiency

- **Optimize discount allocation** by focusing promotions on low-demand periods or new customer acquisition rather than frequent blanket discounts.
- **Monitor low-performing restaurants and items** to reduce promotional spending on partners that do not generate sufficient revenue.
- **Encourage digital payment methods** through small incentives to reduce cash handling costs and improve transaction efficiency.

16.4 Operational Improvement

- **Reduce delivery time variability** by improving delivery agent allocation and optimizing routing in high-demand locations.
- **Implement performance-based incentives for delivery agents** using rating and delivery time metrics to enhance service quality.

- **Balance delivery workload across agents and cities** to prevent delays during peak hours and maintain consistent customer experience.
- **Continuously monitor restaurant preparation and delivery coordination** to minimize order delays and improve customer satisfaction.

17. CONCLUSION

The **Online Food Delivery Business Intelligence & Performance Analysis Using SQL** project successfully demonstrates how structured relational data can be leveraged to generate meaningful business insights and support strategic decision-making. By analyzing customer information, restaurant performance, order transactions, item-level sales, and delivery operations, the project provides a comprehensive view of the platform's operational efficiency and revenue dynamics. The use of SQL enabled effective data extraction, aggregation, and multi-table analysis, highlighting key performance indicators such as net revenue, customer retention patterns, cuisine demand, delivery efficiency, and product performance.

The insights derived from this analysis emphasize the importance of data-driven strategies in improving customer engagement, strengthening restaurant partnerships, optimizing discount utilization, and enhancing delivery operations. Identifying high-performing restaurants, frequently ordered items, and repeat customer behavior provides valuable direction for targeted marketing, menu optimization, and revenue growth initiatives. Additionally, operational findings related to delivery time and agent performance highlight opportunities for improving service quality and customer satisfaction.

Overall, this project reflects the practical application of SQL in solving real-world business problems within a marketplace environment. It showcases strong analytical thinking, database querying proficiency, and the ability to translate raw data into actionable insights. The outcomes of this analysis can support the platform in achieving sustainable growth, improved operational efficiency, and enhanced customer experience, making the project a strong representation of end-to-end business analytics capabilities.

18. FUTURE SCOPE

The current SQL-based analysis establishes a strong foundation for business intelligence; however, the project can be further enhanced by incorporating advanced technologies and scalable analytical solutions to support evolving business needs.

18.1 Automation Possibilities

Future improvements can include automating data extraction, transformation, and reporting processes through scheduled SQL jobs and ETL pipelines. Automated KPI generation and periodic performance reports would reduce manual effort while enabling faster decision-making and continuous business monitoring.

18.2 Integration with BI Tools

The database can be integrated with Business Intelligence tools such as **Microsoft Power BI**, **Tableau**, or **Looker** to develop interactive dashboards. This would allow stakeholders to visualize revenue trends, customer behavior, restaurant performance, and delivery metrics in real time, improving accessibility and strategic planning.

18.3 Real-Time Analytics

Implementing real-time data pipelines can enable live tracking of order volume, delivery performance, and revenue generation. Real-time analytics would help the business respond quickly to operational issues, manage peak demand efficiently, and improve customer experience through proactive decision-making.

18.4 Advanced Modeling Opportunities

The dataset can be extended to support advanced analytical models such as customer segmentation, cohort analysis, demand forecasting, and profitability analysis. Incorporating additional attributes like order status, delivery distance, and customer feedback would further enhance analytical depth and business insights.

18.5 Machine Learning Integration

Future enhancements can include applying machine learning techniques to predict customer churn, recommend personalized food items, forecast demand for specific cuisines, and estimate delivery time more accurately. Integrating predictive models with the existing SQL analytics framework would enable the transition from descriptive analytics to predictive and prescriptive decision-making.